

Pepeyoca: Reconsidering the bioreactor as a bioluminescent living wall for human-bacterial co-inhabitation

Aurélie Mosse, Mette Ramsgaard Thomsen, Vincent Rennie, Daniel Suárez Zamora, Guro Tyse, Martin Tamke

▶ **To cite this version:**

Aurélie Mosse, Mette Ramsgaard Thomsen, Vincent Rennie, Daniel Suárez Zamora, Guro Tyse, et al.. Pepeyoca: Reconsidering the bioreactor as a bioluminescent living wall for human-bacterial co-inhabitation. SIGraDI 2024 -Biodigital intelligent systems, Nov 2024, Barcelona, Spain. ⟨hal-05003052⟩

HAL Id: hal-05003052

https://hal.science/hal-05003052v1

Submitted on 9 Apr 2025

HAL is a multi-disciplinary open access archive for the deposit and dissemination of scientific research documents, whether they are published or not. The documents may come from teaching and research institutions in France or abroad, or from public or private research centers.

L'archive ouverte pluridisciplinaire **HAL**, est destinée au dépôt et à la diffusion de documents scientifiques de niveau recherche, publiés ou non, émanant des établissements d'enseignement et de recherche français ou étrangers, des laboratoires publics ou privés.



Distributed under a Creative Commons CC BY 4.0 - Attribution - International License

sigradi2024
biodigital_intelligent_systems

iBAG
UIC
BCN

Pepeyoca: Reconsidering the bioreactor as a bioluminescent living wall for human-bacterial co-inhabitation

Aurélie Mosse[1], Mette Ramsgaard Thomsen[2], Vincent Rennie[1], Daniel Suárez Zamora[1], Guro Tyse[1], Martin Tamke [2]

[1]Ecole des Arts Décoratifs – PSL, Paris, France, [2]Royal Danish Academy, Copenhagen, Denmark

aurelie.mosse@ensad.fr; mette.thomsen@kglakademi.dk;
vincent.rennie@ensad.fr; guro.tyse@ensad.fr; martin.tamke@kglakademi.dk

Abstract. This paper contributes to the search for novel design principles for an interspecies architecture. Departing from the design and realisation of the installation 'Pepeyoca – light from within': a bio-digitally crafted living wall, we examine emerging design strategies that can entangle human and more-than-human concerns at multiple scales and across distinct life forms. By focusing on bioluminescent bacteria as a living light source and heather as a soil remediator and oxygen producer, the paper presents novel design strategies for circular interdependency in which multiple species directly support and provide foundational conditions for one another. We describe and discuss novel design and fabrication drivers that reject anthropocentric notions of functionalisation instead bringing the bioreactor out of the lab and into a shared living arena. This is done across multiple steps, ideating the role and performance of an interspecies architecture, searching for interdependencies and developing care protocols to steer their balance.

Keywords: Bio-digital architecture, Living architecture, Bioreactor design, Bioluminescence, Co-inhabitation

1 Introduction

1.1 Bioluminescence in design

Bioluminescence relates to the chemical emission of light by living organisms, mainly from the marine environment. Often used as a marker in biology and

medicine, bioluminescence is a metabolic process that can be well controlled in multiple model organisms. Bioluminescence has been explored as a living light source across architecture and design (Estevez, 2016; van Dongen, 2019; Roosegaarde, 2019). Here, key experiments have focused on the living organism as a potential form of sustainable lighting. In prior works, we investigated the emergent design principles for a living architecture by developing advanced digital design chains for 3d modeling, 3d robotically extruding growth medium and simulating the resulting life spans of bioluminescent bacterial cultures (Ramsgaard Thomsen et al., 2022; Tyse et al., 2022). These laboratory experiments cast bioluminescence as a deliberately temporary light source in which the impact of the complex topology of the medium of the bacterial living conditions could be examined. The aim of this further research is to ask how these performances can be brought out of the lab and into a shared interspecies space of co-inhabitation to fully bring forth the value of interdependency and more-than-human design concerns.

1.2 *Photobacterium kishitanii* as bacterial co-inhabitant

The *Pepeyoca* installation (Figure 5) is based on the *Photobacterium kishitanii* bacteria as a source of bioluminescence. This Photobacterium genus lives mostly in symbiosis within the light organs of deep-sea fishes (Ast et al., 2007; Dunlap & Urbanczyk, 2013). The bioluminescence behaviour can be sustained independently of larger metabolic processes in the host organism. This requires specific care and maintenance based on a strong understanding of their biology and environmental needs. *Photobacterium kishitanii* is a rod-shape facultatively aerobic bacteria known to survive a large range of temperatures from 8 to 34°C (Bjornsdottir-Butler et al., 2016). Like most luminescent bacteria, *P. kishitanii* produces a blue-green light in the range 475-500nm. It is generally reported to emit a higher intensity of light than bacteria of the Vibrio genus (Tanet, 2020). *P. kishitanii* can grow both in the presence and absence of oxygen, relying on fermentation when oxygen is lacking. Its growth also depends on a mix of mainly minerals, especially sodium chloride. It particularly thrives in the presence of divalent cations like Mg2+ and Ca2+ (Bjornsdottir-Butler et al., 2016). Bioluminescence and cell growth occur simultaneously from the beginning of the bacterial culture. This is different from the typical behaviour observed in bioluminescent model organisms of the Vibrio genus, where light emission arises only after a specific cell density is reached. However, an absence of bioluminescence does not mean the bacterial colony is no longer alive. Their light intensity is known to reach a maximum peak at the end of their exponential growth phase (about 25 hours) and to diminish over successive cultures (Tanet, 2020, p.73). Additionally, the *P. kishitanii* species can ferment glucose, leading to the creation of unfavorable acidic conditions for their host organisms, and some *P. kishitanii* strains are known to generate toxic levels of histamine (Bjornsdottir-Butler et al., 2016).

1.3 From bioreactor design to bacterial co-inhabitation

The aim of the *Pepeyoca* installation is to design a shared architecture occupied by multiple organisms. As such, it develops methods of design integrating a bioreactor wall structure. Bioreactors ensure optimal conditions of growth and renewal of microbial colonies. They form the backbone of practices in scientific labs and manufacturing plants, where bacteria are used in the crafting of, for instance drugs, drinks or biofabricated materials.

As artifacts, bioreactors are primarily an engineering achievement, in which the interplay of physical, chemical and biological parameters plays a fundamental role in its performance and stability. Bioreactors are also used outside of the context of the wet lab or the biomanufacturing plant by architects and bio-designers to understand the cultural potential and stakes of microbiologic fabrication in everyday life (Parker, 2023; Crawford, 2024). As technical apparatus, bioreactors can take multiple forms but are generally hermetic objects connected to a supporting infrastructure of sensors, control systems, and mechanism, that provide and extract substances and serve a single function: to provide an optimal environment for living cells to grow effectively (Mandenius, 2016). The grown organism is usually cultivated to perform a single task and later disposed when the task is completed. The bioreactor can therefore be understood as a hermetic space underpinned by anthropocentric thinking and driven by principles of isolation and sterility.

With *Pepeyoca,* we are interested in exploring the co-habitation of bacteria with humans. We do this through a conceptual translocation of the bioreactor from the lab into the world of humans. In doing so, we adapt the technologies of the bioreactor to create a shared spatial whole. This integration occurs at multiple levels, where the usual technological overhead is reduced, and a more immediate interaction is introduced. Firstly, at a material level, where typical bioreactor materials such as steel and glass containers are substituted with more versatile materials that expose the behaviour of the bacteria in the form of 3d-printed polycarbonate filament and silicone sheeting. Secondly, at a technological level in which systems are minimised and open platform programmable tools are used to steer the water flow in the bacterial receptacles. Finally, and more profoundly, at a relational level, the purpose and performance of the bacterial receptacles are designed not just for optimal light performance but also for colony growth, longevity and survival. As such, the design aims in the *Pepeyoca* project move away from viewing the bioreactor as a fully controlled environment and instead develop the ambition to create a symbiotic environment as an integral part of architectural materiality.

2 Research by design methodology: bio-design as architectural probes

Pepeyoca is the outcome of a design-led and interdisciplinary approach to research informed by the creative practices of architecture, bio-design and digital crafting. By questioning the concepts, crafts and interactions through which bacterial bioluminescence can become part of the built environment, the research takes part in an understanding of bio-design as the creation of living systems or artefacts in which preserving the bacterial vitality is central (Karana et al., 2023). In *Pepeyoca,* we adopt a bottom-up design process driven by the lifecycle of *P. kishitanii*, underpinned by circular and more-than-human design principles (Puig de la Bellacasa, 2017; Charter, 2019). More specifically, we are working with biosafety level 1 bioluminescent bacteria *P. Kishitanii* isolated from marine fish. This means we are exclusively working with an existing biological system encountered in the environment. Following a design probes tradition (Mosse, 2015); prototyping remains the primary means to develop and answer both the research question and design criteria. As an installation developed for a six-month public exhibition, *Pepeyoca* seeks to understand the needs and conditions of care that co-inhabiting a shared environment with bioluminescent bacteria entails.

The architectural inspiration for *Pepeyoca* is Peter Cook's Veg House drawing set from 1996 to 2001 (2016). Drawn across six growth stages, the speculative and deliberately ambiguous project examines how architecture can integrate living organisms and how they can evolve into a novel framework for living. Our interest in 'Veg House' lies in the evolving language that replaces core architectural axioms such as boundary, ground and site. Instead, new relationships are formed, interweaving and enfolding the organic and the built.

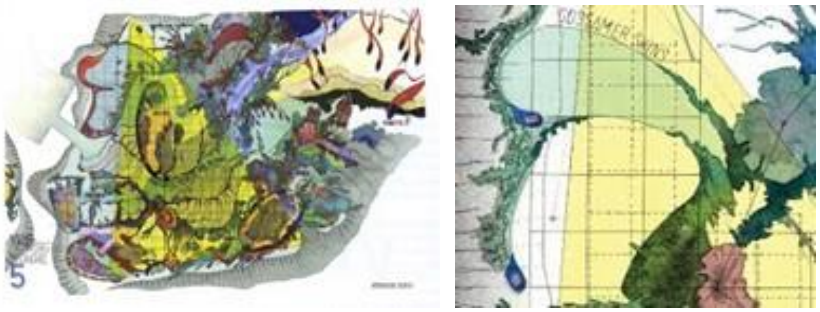


Figure 1. Peter Cook's Veg House Stage 4 and detail. Source: Cook, 2016.

Here, organic matter becomes the living boundaries of architectural space, and new hybrid tropes integrate layers of organic and artificial materiality (Figure 1). Specifically, in Stage 4, we interpret the emergence of "Gossamer Skins" as semipermeable membranes supporting organic growth, membranes

that let small molecules like oxygen and water pass through, but not larger particles that can cause contamination. In *Pepeyoca* this inspiration is translated into a two-sided wall fragment. Borrowing its name from the Nahuatl language, in which *Pepeyoca* evokes an inner shimmer or lively sparkle (Molina, 1571; Whorf, 1928), the interior wall side integrates the bioreactor vessels and their infrastructure and the exterior side interfaces a heathered landscape that participates in the circular interaction between species. The wall fragment is made as a chunk model (Kolatan, 2020), a compact part model that describes the principle of architectural interactions.

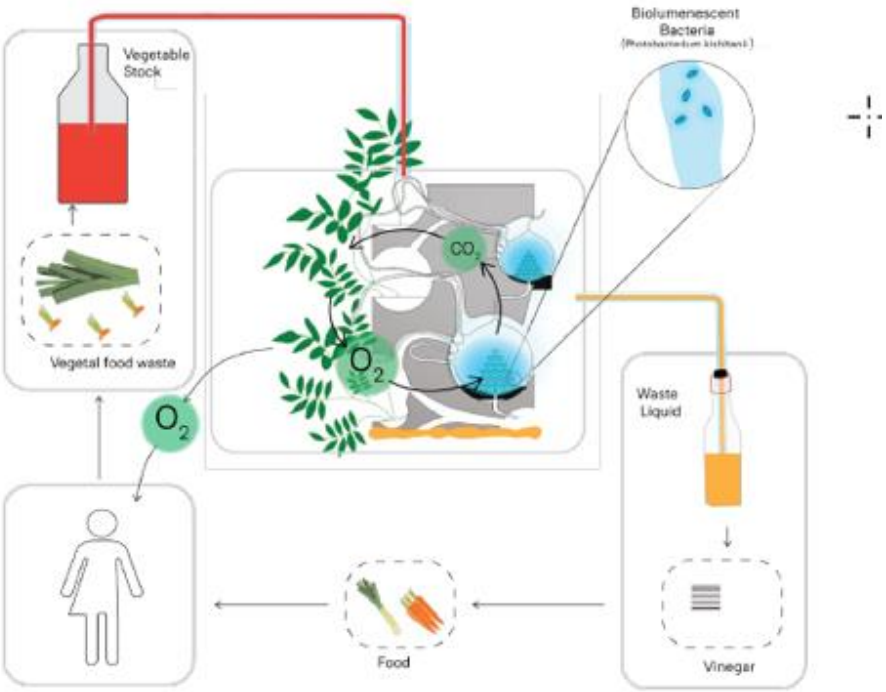


Figure 2. Pepeyoca's inter-species metabolic principles. Source: Soft Matters, 2023

2.1 Pepeyoca as a circular living wall

The *Pepeyoca* installation entangles three scales and distinct life forms. It builds a circular logic in which the effluents of one species nurtures the others (Figure 2). By creating an ecological setting in which bacterial nutrition passes through the system, feeding *P. kishitanii* while bacterial waste and toxins become the setting for a heathered landscape, requiring acidic soil to thrive. In turn, the heather generates oxygen which enables both the light performance of the bacteria and the oxygenation of the space, which benefits both humans and the bacterial illumination. The bacterial efflux could also potentially become

a source of acetic acid, namely vinegar, for human consumption. Waste from human alimentation could equally be used as plant fertilisers and bacterial nutrition to nurture these loops. These entanglements create an ecological setting, of which the materialisation is described and discussed in the subsequent sections.

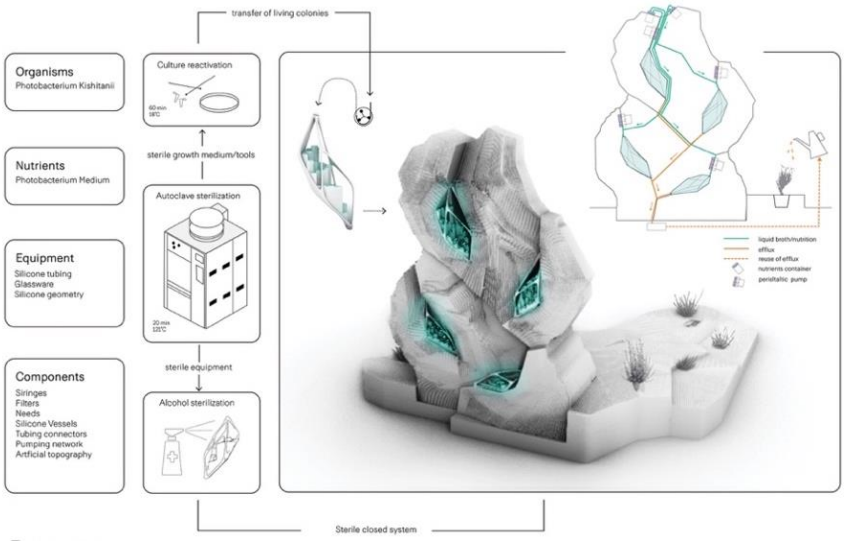


Figure 3. Modelling of Pepeyoca and its maintenance system. Source: Daniel Suárez, 2023.

2.2 Pepeyoca as an integrated bioreactor

Design overview

In *Pepeyoca*, we develop an integrated bioreactor that is distributed across four light vessels. The vessels are nestled into a wall structure, creating a continual light source on one side and the heathered landscape on the other. The wall itself is made of seven concrete blocks, which are fabricated in CNC-milled aerated concrete. The surfaces are purposely highly detailed with a patterned surface treatment to reflect and give depth to the bioluminescent light.

Starting from a diagrammatic understanding of the system's metabolism (Figure 2), we develop the system at two parallel levels: firstly, a biological level, and secondly, a design and infrastructural one. In contrast to a consecutive approach, where small-scale experiments in the lab are later developed into larger scales, this method allows us to handle the interdependency of design and biological findings to guide the overall development.

Bio-system development

The aim of this design level is to develop protocols and infrastructures that allow for the maintenance of bacterial life over a long period of time. Through small-

scale lab tests, we investigate how nutrition can be given in an automated and controlled way. While traditional bioreactors employ complex sensor and feeding systems (Sonnleitner, 2006), our tests explore gravity-based nutrient drip systems to facilitate nutrient flow and toxin evacuation.

In the first tests, we use a translucent silicone base - cast from a 3D printed counter-mould - with a simple spiral geometry combined with a commercially available drip feed system to create a fluid flow over the surface (Figure 4). We use a state-of-the-art marine broth as nutrition (Merck, n.d.), and the freeze-dried bacteria are reactivated and pre-grown in the liquid broth at 18°C before inoculation into the system after 24 hours.



Figure 4: Setup of the experiment in a lab incubator. Drip feeding system on left, substrate in sealed glass vessel in the middle (left). Documentation of bioluminescence with SLR camera after 3 days of culture (right). Source: Soft Matters, 2023.

Design development

The *Pepeyoca* installation is spatially organised as a network of four light vessels nested in the wall structure at various heights and connected in three parallel circuits (Figure 3). The circuits are fed at the top of the wall, where the nutrition tank is positioned while at the bottom, the effluents are evacuated towards the wall's other side where, the heathered landscape is placed. The circuit connections are ensured by flexible tubing and aseptic connectors. Four digitally controlled peristaltic pumps located just before each light vessel head allow nutrition to flow control while gravity-based valves permit liquid evacuation. The nutritive broth is programmed to flow at a rate of 2.5ml per hour, equivalent to 0.04ml per minute, to ensure a soft, progressive renewal of the nutritive liquid broth.

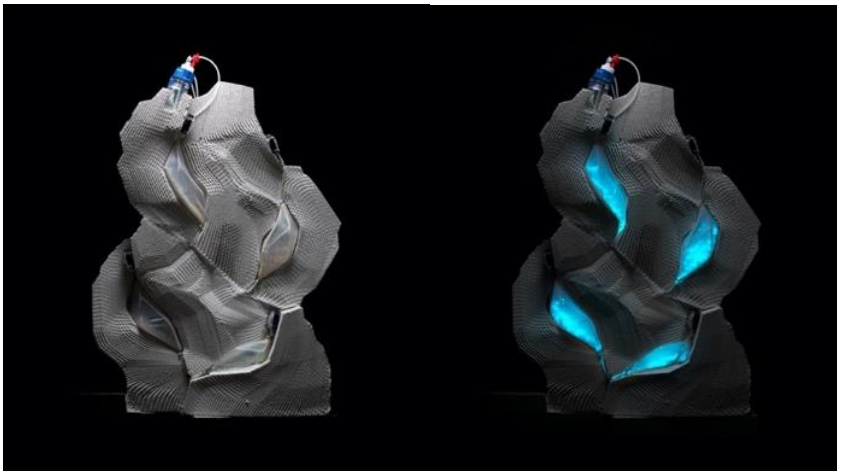


Figure 5. Pepeyoca inner side view, without (left) and with bioluminescence (right). Source: Anders Ingvartsen, 2024**.**

2.3 Pepeyoca light 's vessels

In all there are four light vessels in *Pepeyoca* (Figure 5). Each vessel is formed as a unique topology, precisely adjusted to fit within the programmed interstices of the chunk model into which they are inserted. The light vessel design consists of a 3D-printed skeleton wrapped with a developable silicone skin (Figure 6). This structure holds an interior terraced surface of shallow basins. This landscape of pools maximises the bioluminescent emitting surface, increasing the light levels for humans. It builds on earlier studies in which we shown that *P.kishitanii* thrives not in fluids, but best on the surface of shallow areas (Ramsgaard Thomsen et al., 2021; Mosse et al., 2023).

The terraced design is 3D printed in polycarbonate, an impenetrable thermoplastic. The shallow basins are filled with marine broth inoculated with the bacteria (Figure 7). The vertical terracing of the design allows the fluids to flow through the basins from the highest point to the lowest. This creates a flushing of the system in which the nutrition dripping in the topmost basin flows through all lower basins, removing at the same time the bacteria's slightly acidic effluent to the external heather.

The exoskeleton is clad in a transparent silicone skin. Learning from inflatable bioreactors (Dufour et al., 2024), silicone is chosen as it is hermetically closed for contaminants while permeable for light and oxygen. This allows the bacteria to have continual airflow which is essential for its bioluminescence behaviour. Silicone is also more amenable to our production facilities than glass containers and allows a higher degree of precision and geometrical freedom.

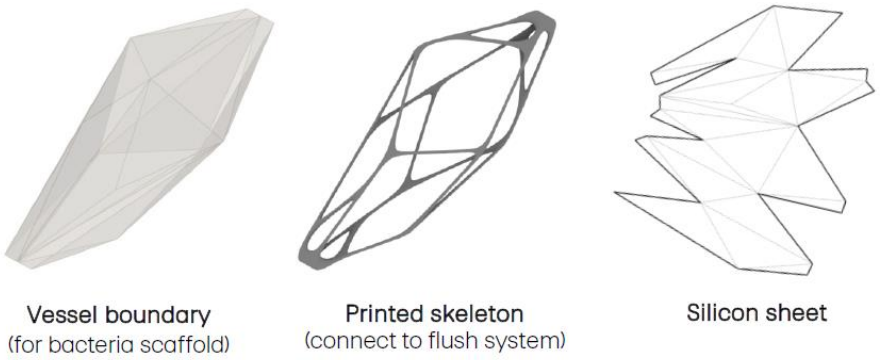


Figure 6. Generation of the vessel: low poly model (left), exoskeleton as isosurface from the vertices, silicone sheet as 2 dimensional unroll. The base geometry of the skin is unrolled from the base polygon of the vessel and adapted according to measured stretch and behaviour of the skin. Source: CITA, 2023.

However, during the design process we found that 500-micron silicone film was needed for manufacturing reasons. Our calculation shows that the permeability of this material thickness is not sufficient for the necessary oxygen flow. To support further oxygen flow, we therefore include HPE filters at the aeration of the nutrition bottles.

The efflux is gathered at the bottom of the installation and fed to the heathered landscape nested between the concrete blocks. The heather thrives on its acidity and supports its production of oxygen. The two sides of the wall structure are therefore symbiotically connected as multiple more-than-human species interact to create an ecological whole.

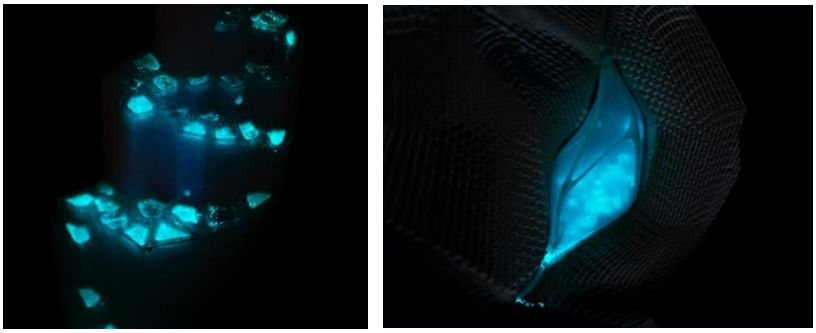


Figure 7. Details of the algorithmically created terraced illuminated substrate (left). Same substrates as inserted into its vessel (right) Source: Daniel Suárez, 2023 (left) Anders Ingvartsen, 2024 (right).

3 Results & discussion

3.1 Light performance

The bioluminescence is an indicator of the bacteria's vitality. Within the lab environment, we observe light emission for up to 10 days while the light performance of *Pepeyoca* in the exhibition context is reduced to 3 days, with a peak of light 24 hours after inoculation. The core reason for the shorter life expectancy of the bacteria in the installation setup is due to the strong fluctuations in temperature in the gallery space. Here, we observed deviations of ± 10 degrees Celsius from the ideal temperature for the bacteria. Secondary reasons could include contamination. While we observed other bacterial species (black mould) in one of the four vessels, we expect that some foreign bacteria settled in other parts of the system.

3.2 Biosystem development

While the flow system necessitates calibration and adjustment of the circulation tubes to interface well with the capacity of the small-scale peristaltic pumps, the nutrition system in our integrated bioreactor is functioning well. However, we that the effluent pump was not sufficiently strong and did not seal the effluent tube. This meant that fluid from higher-positioned effluence tubes could run down into the lowest vessel, causing temporary flooding.

The flow within the vessels also worked well. However, the minute stepping of the pools required very precise positioning of the vessels as they are nested between the aerated concrete blocks. Within the exhibition context, this turned out to be a complex task as the tolerances in fabrication and installation of the chunk wall pieces and, the vessels themselves made this precision difficult to achieve. In future iterations a stronger reference point would simplify the process of installation as well as the continual re-inoculation.

3.3 Bioreactor as an architectural materiality

Pepeyoca brings together different architectural performances: light performance, the reflection and the nested landscape, supported through various design processes and material fabrications.

Pepeyoca exhibits a spatially extended, more refined, and lively quality of illumination compared to prior results (Ramsgaard Thomsen et al., 2021), due to its distributed system and terraced substrates. Light reflections induced by the silicone skin and surfaced concrete further enhance this bioluminescent experience (Figure 7). While we did not succeed, in gallery conditions, to maintain more than a 3-day light performance, we assume - based on Tanet's results and an improved inoculation, monitoring and maintenance protocol-, to be able to sustain bioluminescence over a 1-month period in the next iteration (2020).

Overall, this attention to light performance and concerns for its integration as an essential part of a shared living wall, through which three species can co-inhabit, allows the bioreactor to acquire an architectural expression that radically departs from its traditional appreciation as an engineered artefact.

4 Conclusion

In *Pepeyoca* we deliberately seek to design and integrate a multispecies environment in which species of very different kinds and temporalities can interact as a circular living wall. As a chunk model, the *Pepeyoca* installation shows a fragment of a larger, undefined whole. It points to the systemic interrelationships that can be connected and supported, as well as a sense of performance through ideas of light performance and oxygen production. However, our aim in *Pepeyoca* is not to functionalise bioluminescence but to probe how architectural design can integrate bacterial luminescence within a virtuous loop of interactions between three life scales and forms, where one draws its energy from the waste of the others. These new interdependencies transform relationships among living beings. Bacteria become a source of light, their effluent nurtures plants, while human co-inhabitants become their caretakers. This shift has a range of implications for technology as well as inhabitants. While the later needs to take on the role of caretaker of the bacteria, engaging in protocols of caretaking and protection of bacteria from e.g. other bacteria, that wants to take over the space, the technology needs as well to adapt. Here the bioreactor literally needs to open up, as it is not only the space for bacteria to grow, but as well a source of light. As such it needs to change materiality and integrate into the space of humans.

Acknowledgements. *Pepeyoca* is an outcome of *ImpressioVivo,* a research project supported by Agence Nationale de la Recherche (ANR-21-CE43-0023-01) and of the European Research Council funded project: Ecometabolistic Architecture (No.101019693). We thank the extended team for design, making and caring: C. S. Lim, M. Waschek, A. Sato, S. Akbari, T. Rayne.

References

Ast, J. C., Cleenwerck, I., Engelbeen, K., Urbanczyk, H., Thompson, F. L., De Vos, P., & Dunlap, P. V. (2007). Photobacterium kishitanii sp. Nov., a luminous marine bacterium symbiotic with deep-sea fishes. *International Journal of Systematic and Evolutionary Microbiology*, *57*(9), 2073–2078. https://doi.org/10.1099/ijs.0.65153-0

Bjornsdottir-Butler, K., McCarthy, S. A., Dunlap, P. V., & Benner, R. A. (2016). Photobacterium angustum and Photobacterium kishitanii, Psychrotrophic High-Level Histamine-Producing Bacteria Indigenous to Tuna. *Applied and*

Environmental Microbiology, *82*(7), 2167–2176. https://doi.org/10.1128/AEM.02833-15

Charter, M. (Ed.). (2019). *Designing for the circular economy.* Routledge, Taylor & Francis.

Cook, P. (2016). *Architecture workbook: Design through motive.* Wiley.

Crawford, A. (2024). *Designer's guide to lab practice*. Routledge.

Dufour, A., Essayan, L., Thomann, C., Petiot, E., Gay, I., Barbaroux, M., & Marquette, C. (2024). Confined bioprinting and culture in inflatable bioreactor for the sterile bioproduction of tissues and organs. *Scientific Reports,* 14(1), 11003. https://doi.org/10.1038/s41598-024-60382-2

Dunlap, P., V., & Urbanczyk, H. (2013). Luminous bacteria. In *The Prokaryotes—Prokaryotic Physiology and Biochemistry* (4th ed., pp. 495–528). Springer-Verlage.

Estevez, A. T. (2016). *Towards Genetic Posthuman Frontiers in Architecture & Design*. 450–459. https://doi.org/10.52842/conf.acadia.2016.450

Karana, E., McQuillan, H., Rognoli, V., & Giaccardi, E. (2023). Living artefacts for regenerative ecologies. *Research Directions: Biotechnology Design*, 1, e16. https://doi.org/10.1017/btd.2023.10

Kolatan, F. (2020). The Chunk Model. Log No. 50, *Model Behavior*, Fall 2020, 71–74. https://www.anycorp.com/store/log50

Mandenius, C.-F. (2016). Chapter 1 Challenges for Bioreactor Design and Operation. In C.-F. Mandenius (Ed.), *Bioreactors: Design, operation and novel applications*. Wiley-VCH Verlag GmbH & Co. KGaA.

Merck. (n.d.). *Photobacterium Broth* [Commercial]. Sigma Aaldrich. Retrieved July 18, 2024, from https://www.sigmaaldrich.com/

Molina, A. de O. F. M. (1571). *Vocabulario en lengua castellana y mexica*na (en casa de A. de Spinosa, Ed.; Vol. 2). en casa de Antonio de Spinosa. https://bibliotecadigital.aecid.es/bibliodig/es/consulta/registro.do?id=132

Mosse, A. (2015). *Gossamer Timescapes: Designing self-actuated textiles for the home* [PhD thesis]. Royal Danish Academy.

Mosse, A., Tyse, G., Tamke, M., & Ramsgaard Thomsen, M. (2023). Imprimer la Lumière: Bacterial Luminescence as a 3D-Printed Spiral Micro-Architecture. *.able Journal*. https://able-journal.org/imprimer-la-lumiere

Parker, B. (2023). How can we design robust manufacturing metabolisms and reconsider the bioreactor? *Research Directions: Biotechnology Design*, *1*, e3. https://doi.org/10.1017/btd.2022.4

Puig de la Bellacasa, M. (2017). *Matters of care: Speculative ethics in more than human worlds*. University of Minnesota Press.

Ramsgaard Thomsen, M., Tamke, M., Mosse, A., Sieder-Semlitsch, J., Bradshaw, H., Buchwald, E. F., & Mosshammer, M. (2022). Imprimer La Lumiere – 3D Printing Bioluminescence for Architectural Materiality. In P. F. Yuan, H. Chai, C. Yan, & N.

Leach (Eds.), *Proceedings of the 2021 DigitalFUTURES* (pp. 305–315). Springer Singapore.

Ramsgaard Thomsen, M., Tamke, M., Mosse, A., & Tyse, Guro. (2021). *Designed Substrates For Living Architecture Performance—Imprimer La Lumière*. CEES 2021 – Construction, Energy Environment & Sustainability, University of Coimbra, Portugal. https://cees2021.uc.pt/projectos/cees2020/atas/pdfs/ID169.pdf

Roosegaarde, D. (2019). Glowing Nature. In M.-A. Brayer & O. Zeitoun (Eds.), *La fabrique du vivant: Mutations, créations* (pp. 206–207). HYX.

Sonnleitner, B. (2006). Measurement, monitoring, modelling and control. In C. Ratledge & B. Kristiansen (Eds.), *Basic Biotechnology* (3rd ed., pp. 251–270). Cambridge University Press. https://doi.org/10.1017/CBO9780511802409.012

Tanet, L. (2020). *La bioluminescence bactérienne, des océans à Biolum-Archi: Un phénomène biologique fascinant au service d'un modèle d'habitat durable méditerranéen?* [PhD Thesis, Aix-Marseille Université].

Tyse, G., Tamke, M., Ramsgaard Thomsen, M., & Mosse, A. F. (2022). Bioluminescent micro-architectures: Planning design in time, an eco-metabolistic approach to biodesign. *Architecture, Structures and Construction*, *2*, 471–479. https://doi.org/10.1007/s44150-022-00038-9

van Dongen, T. (2019). *Ambio* [Graphic]. https://teresavandongen.com/Ambio

Whorf, B. L. (1928). *Investigations in Aztec linguistics and Toltec history*: *Part II, the phenomenon of oligosynthesis in Nahuatl or Aztec* (1956th ed.). University of Chicago Library. https://www.lib.uchicago.edu/mca/mca-07-043.pdf